

Honeywell Aerospace — Software Engineer

10-Day Coding Interview Prep Plan (Python, Pure Software/Backend)

Day 1: Arrays & Strings

- Two Sum
- 3Sum
- Product of Array Except Self
- Longest Substring Without Repeating Characters
- Rotate Array
- Kadane's Algorithm (Maximum Subarray)
- Minimum Window Substring (sliding window)

Day 2: Linked Lists

- Reverse Linked List (iterative + recursive)
- Detect & Remove Cycle (Floyd's Algorithm)
- Merge K Sorted Lists
- Add Two Numbers (linked list representation)
- Clone a Linked List with Random Pointer

Day 3: Trees & BSTs

- Validate BST
- Lowest Common Ancestor (BST and Binary Tree)
- Level Order Traversal
- Zigzag Level Order Traversal
- Diameter of Binary Tree
- Maximum Path Sum
- Serialize / Deserialize Binary Tree

Note: Strong existing area — treat as a speed refresh.

Day 4: Stacks, Queues & Heaps

- Valid Parentheses
- Min Stack
- Next Greater Element
- Largest Rectangle in Histogram
- Kth Largest Element in an Array
- Top K Frequent Elements
- Implement a Queue using Two Stacks

Day 5: Recursion & Backtracking

- Subsets

- Permutations
- Combination Sum
- N-Queens
- Sudoku Solver
- Word Search (grid)

Day 6: Graphs

- BFS / DFS from scratch (adjacency list)
- Number of Islands
- Clone Graph
- Topological Sort (Course Schedule)
- Dijkstra's Algorithm (shortest path)

Day 7: Dynamic Programming

- Fibonacci: memoization to tabulation progression
- 0/1 Knapsack
- Coin Change
- Longest Common Subsequence
- Edit Distance
- House Robber / Climbing Stairs variants

Day 8: Bit Manipulation & OOP Design

- Single Number
- Count Set Bits
- Power of Two
- XOR-based tricks
- Design a state machine (class-based, using Enum for states)
- OOP design problem: Parking Lot / Elevator System / Library System
- Practice clean typed Python: type hints, dataclasses

Day 9: Mock Round #1 + Review

- Timed mock: 2 medium + 1 hard problem, 60-75 minutes
- Redo any problems from Days 1-8 you got stuck on
- Review time/space complexity notes for each pattern covered

Day 10: Mock Round #2 + Final Review

- Second timed mock round, same format as Day 9
- Final pass over weak patterns only — no new topics
- Light day — rest well before the interview

Priority note: If any day runs short, protect Arrays/Strings, Trees, and Dynamic Programming — these patterns show up most often in actual rounds. CS fundamentals and SQL are being prepared separately, so this plan is coding/DSA only.